

Samalka Anandagoda, Ph.D.

LinkedIn: <https://www.linkedin.com/in/samalka-anandagoda/>

GitHub: <https://github.com/samalkapiy>

Website: <https://samalkapiy.github.io/>

SUMMARY

- Data scientist with a research background and 8+ years of experience analyzing large-scale datasets, developing statistical and machine learning models. Experienced in designing, developing and deploying end-to-end machine learning solutions on AWS cloud services using Docker and FastAPI.
- Proficient in designing and implementing A/B testing, conducting statistical analysis and hypothesis testing.
- Delivered machine learning solutions with measurable business impact, including a credit card fraud detection model.

TECHNICAL PROJECTS

- Designed and implemented A/B testing to measure the impact of advertising campaign on user conversion rates, including power analysis, hypothesis testing, bootstrap resampling to quantify statistical significance and treatment effects.
- Credit Card Fraud Detection model built using XGBoost achieving ROC-AUC score of 0.99. Net monthly savings from utilizing the model: 58,638 dollars.
- Space-X Launch success prediction model with XGBoost (cross validation score: 0.89) leading to a 51 million dollars in savings per launch by accurately predicting the outcome of the booster rocket.

Full list containing the slides, codes used for the projects can be found at my website: <https://samalkapiy.github.io/projects/>

RELEVANT RESEARCH AND WORK EXPERIENCE

Aug 2024 - Present	Research Fellow/Visiting Assistant Professor (Physics/Data analysis)	Washington College, Maryland
	- Developing an NLP-driven machine learning pipeline that converts large-scale unstructured scientific reports into validated, machine-readable datasets for statistical analysis and predictive modeling. - Developed and led a hands-on data analysis and scientific computing course focused on exploratory data analysis (EDA), data wrangling, visualization, regression, statistical modeling, and optimization. Mentored students in designing end-to-end analytical workflows, applying Python-based modeling techniques, and communicating data-driven insights through independent projects.	
May 2022 - Jan 2023	Research Assistant	Los Alamos National Laboratory, NM
	- Developed and optimized scalable Monte Carlo simulation models in C++, leveraging parallel computing techniques to accelerate large-scale computational modeling. Automated HPC workflows using bash scripting and job scheduling, enabling efficient execution of large-scale computational experiments.	
Aug 2016- Aug 2023/	Astrophysics Researcher/ Volunteer research fellow	Clemson University, SC
	- Second developer of a Github hosted software release : https://github.com/adesai90/SNuGGY - Designed data analysis pipelines using Python and SQL to process large-scale astrophysical datasets. Applied statistical modeling, maximum likelihood estimation, and data visualization to develop predictive models and solve complex analytical problems, utilizing high-performance computing (HPC) infrastructure to optimize computational performance.	
Aug 2023- Aug 2024		

SKILLS

Tools	Python (PyTorch, SciKit, pandas, matplotlib, numpy, scipy, seaborn), C++, SQL, Jupyter Notebooks, Git, Tableau, AWS, Docker
Concepts	Statistics, hypothesis testing, Machine Learning, Clustering algorithms, Regression and Classification, Data modeling, ETL pipelines, Git version control, High performance computing.
Analytics	Business Intelligence (BI), Dashboard Development, Data Visualization, Reporting.

EDUCATION

Aug 2019 - Aug 2023	Ph.D. in Astronomy and Astrophysics	Clemson University, SC
Aug 2016 - Aug 2019	MS in Astronomy and Astrophysics	Clemson University, SC
Nov 2011 - Nov 2015	Bachelor of Science in Physics Major: Physics, Minor: Computer Science and Mathematics	University of Peradeniya, Sri Lanka